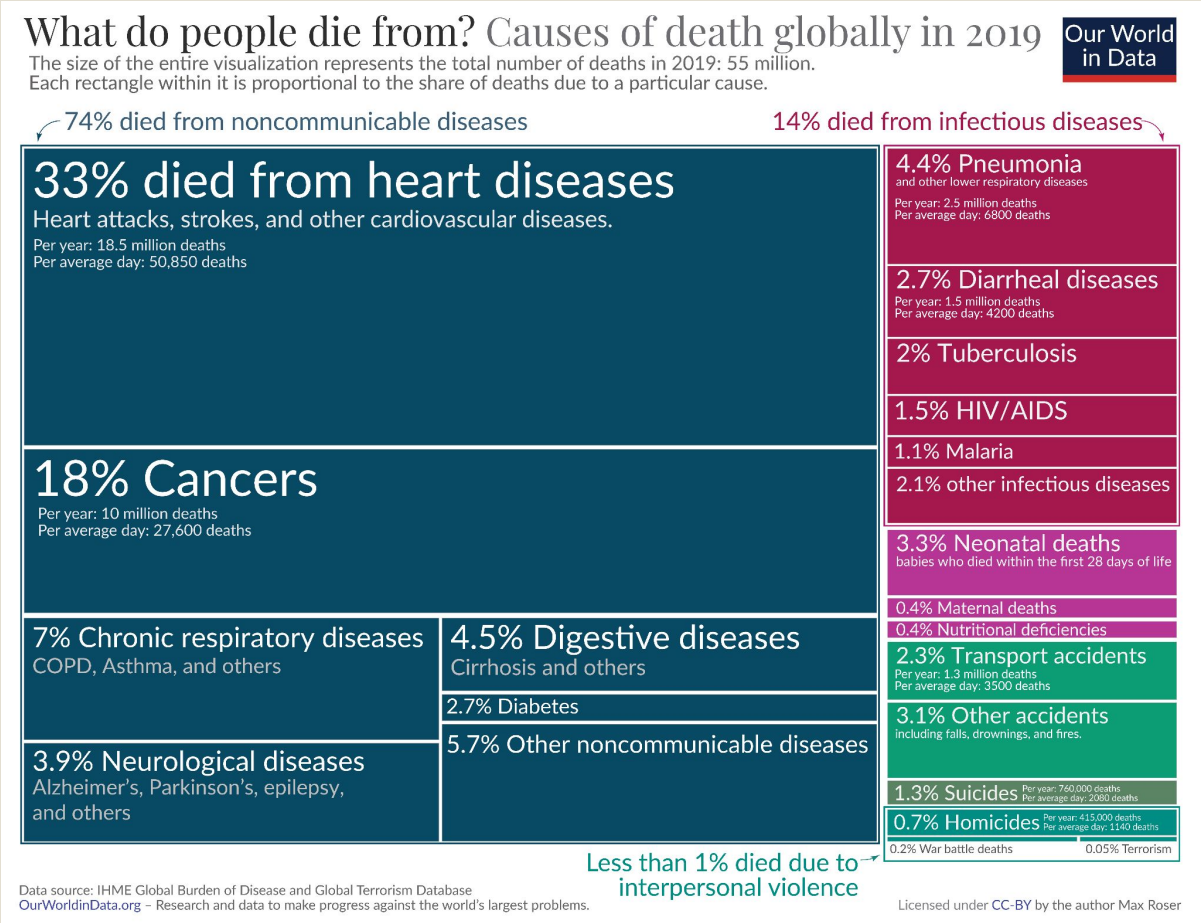
HEART DISEASE PREDICTION MODELING

GROUP 6

Introduction



Source:https://ourworldindata.org/causes-of-death

- disease accounts for **33% of global deaths**.
- Early detection can improve outcomes through timely intervention.
- This project uses medical data to predict heart disease risk, helping identify high-risk individuals for proactive healthcare.Heart

Objective

Identify Key Factors:

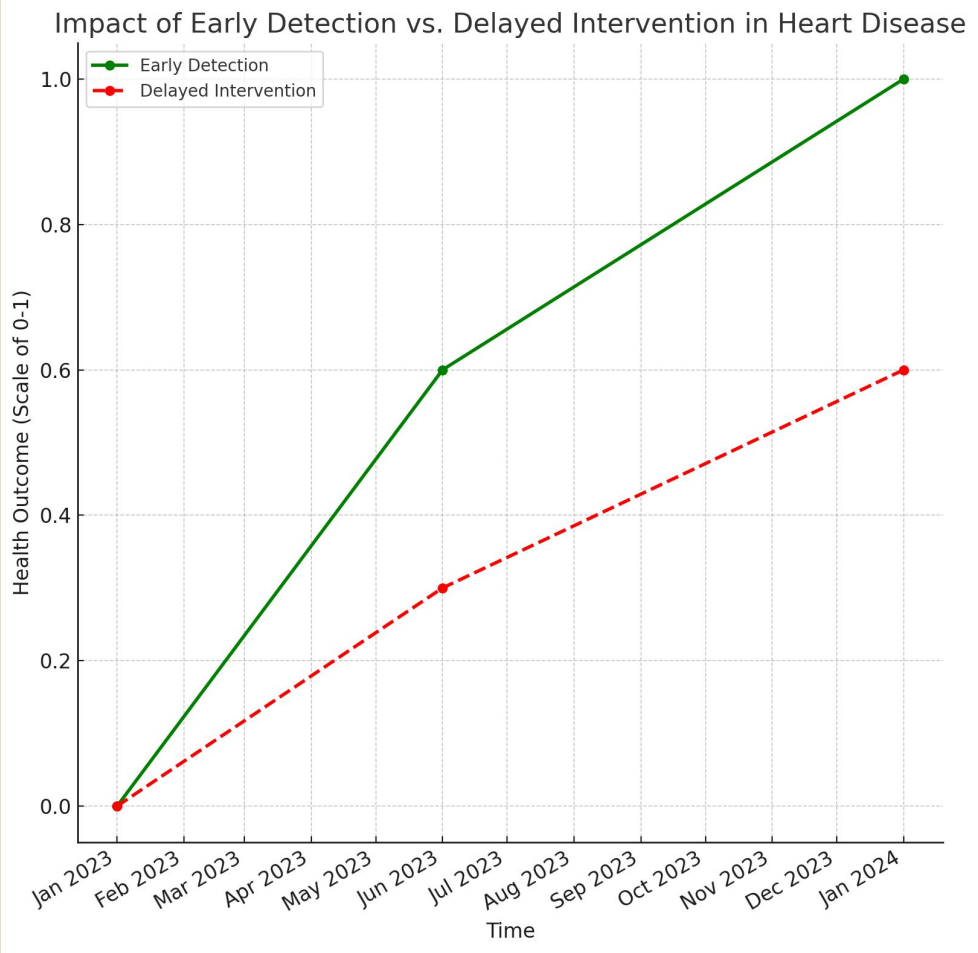
- Analyze medical and demographic data to pinpoint major contributors to heart disease risk (e.g., age, cholesterol, blood pressure).

Develop a Predictive Model:

- Build and evaluate classification models (Logistic Regression, Decision Trees, Random Forest, LDA, QDA) for accurate heart disease prediction.

Impact of Early Detection:

- Explore how early detection improves outcomes and compare the effects of early vs. delayed intervention on patient care.



Source: AI Generated

Business Problem

Can we develop a reliable machine learning model to predict the presence of heart disease, enabling healthcare providers to make more accurate decisions and improve early detection?

Significance:

- **Early Detection Saves Lives:** Timely identification reduces mortality rates and improves patient outcomes.
- **Enables Personalized Treatment:** Provides tailored care, ensuring more effective treatment strategies.
- **Improves Diagnostic Accuracy:** Supports healthcare professionals in making accurate, data-driven diagnoses.
- **Reduces Unnecessary Procedures:** Lowers false positives, minimizing the need for excessive tests and procedures.

Data Overview

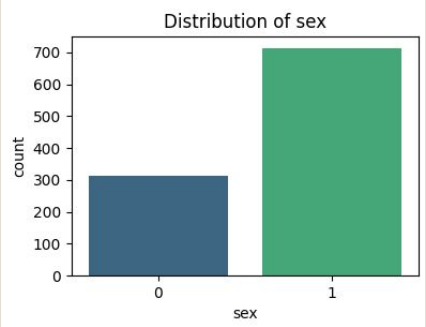
Source: The dataset is publicly available on Kaggle.
Heart Disease Dataset: Kaggle Link

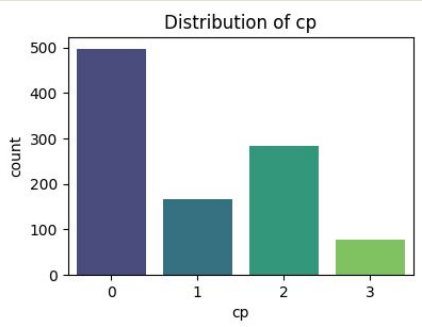
Features: The dataset consists of 14 key medical and demographic features, which are crucial for predicting the presence of heart disease.

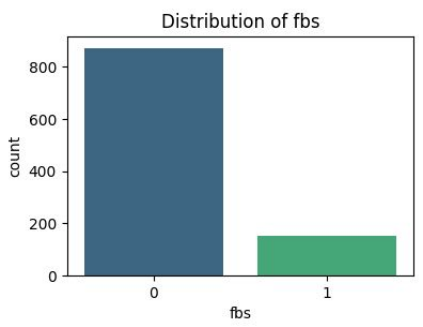
- **Categorical Variables:** sex, cp, fbs, restecg, exang, slope, ca, thal, target
- **Continuous Variables:** age, trestbps, chol, thalach, oldpeak

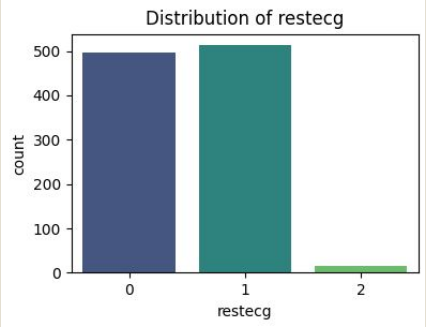
Size: 1,025 observations with 13 columns of data.

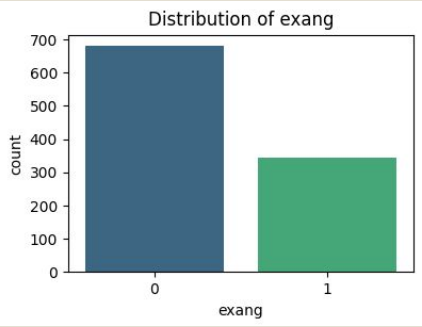
Categorical Variables

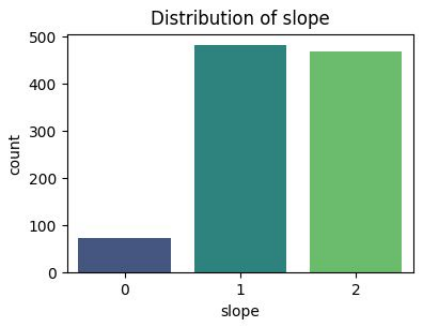


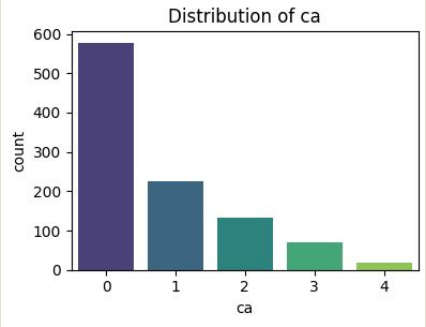


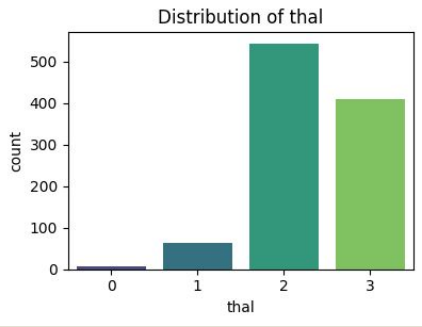


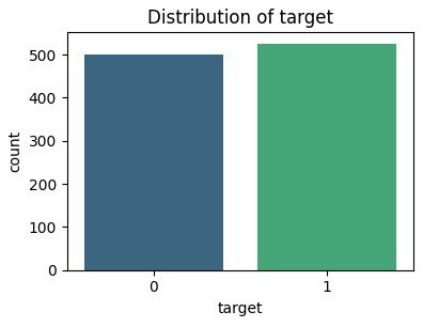






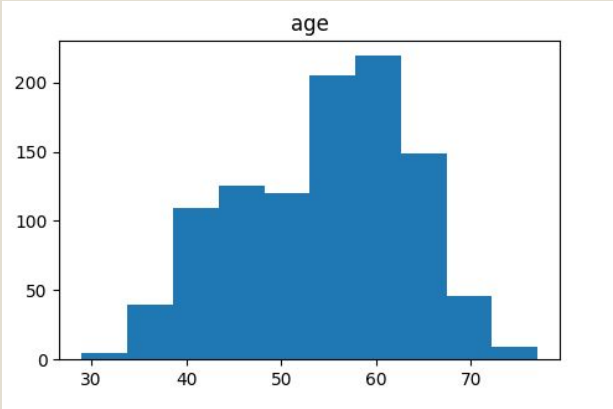




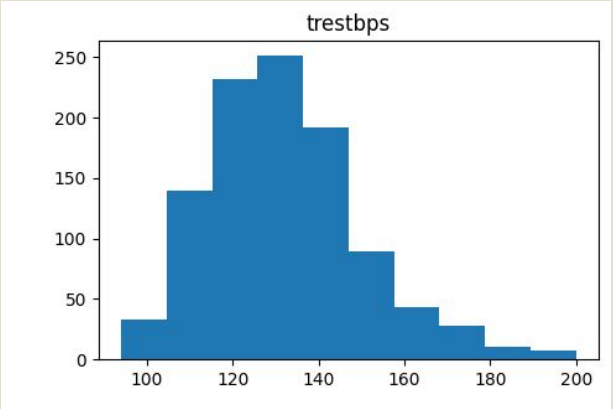


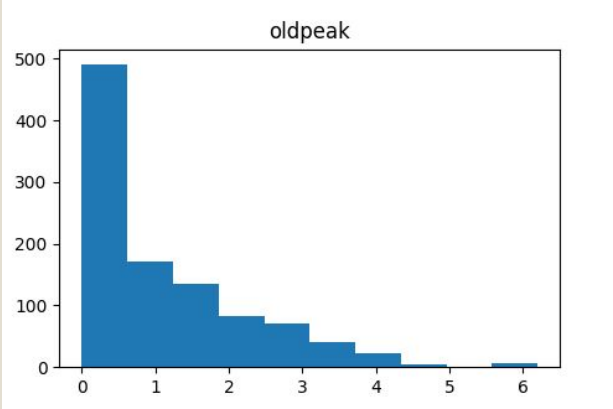
Variable	Description
sex	Gender (1 = Male, 2 = Female)
cp	Chest Pain Type (0 = Typical Angina, 1 = Atypical Angina, 2 = Non-anginal Pain, 3 = Asymptomatic)
fbs	Fasting Blood Sugar (1 = >120 mg/dL, 0 = <=120 mg/dL)
restecg	Resting ECG (0 = Normal, 1 = ST-T Wave Abnormality, 2 = LVH)
exang	Exercise-Induced Angina (1 = Yes, 0 = No)
slope	Slope of ST Segment (0 = Upsloping, 1 = Flat, 2 = Downsloping)
ca	Number of Major Vessels (0–3)
thal	Thalassemia (1 = Normal, 2 = Fixed Defect, 3 = Reversible Defect)
target	Heart Disease Presence (1 = Yes, 0 = No)

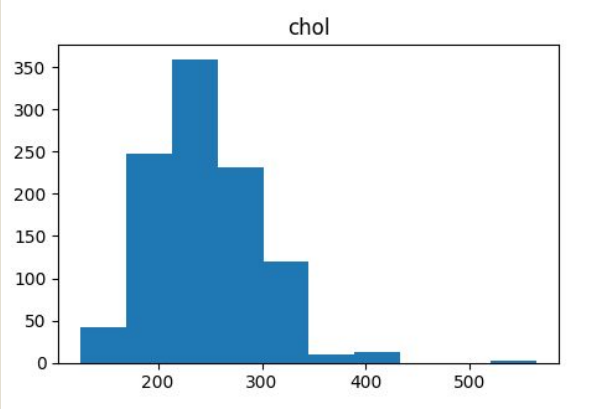
Continuous Variables - Numerical

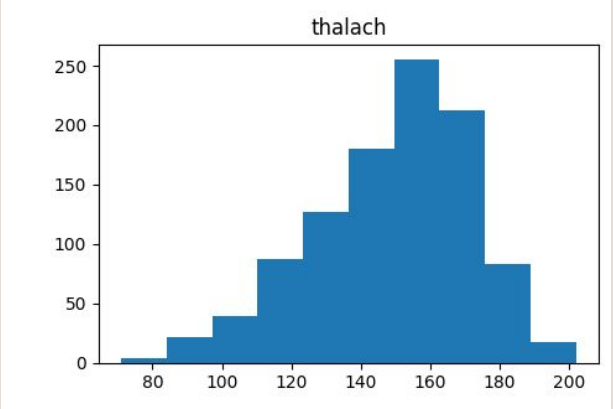


Variable	Description
age	Age of the patient
trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dL)
thalach	Max heart rate achieved
oldpeak	ST depression induced by exercise









PREPROCESSING

Summary Statistics

1.Outliers Detected:

- Cholesterol values up to 564 suggest extreme outliers (75th percentile = 275).
- Resting BP max of 200 may indicate hypertension or data errors.
- Oldpeak max of 6.2 suggests extreme ST segment depression.

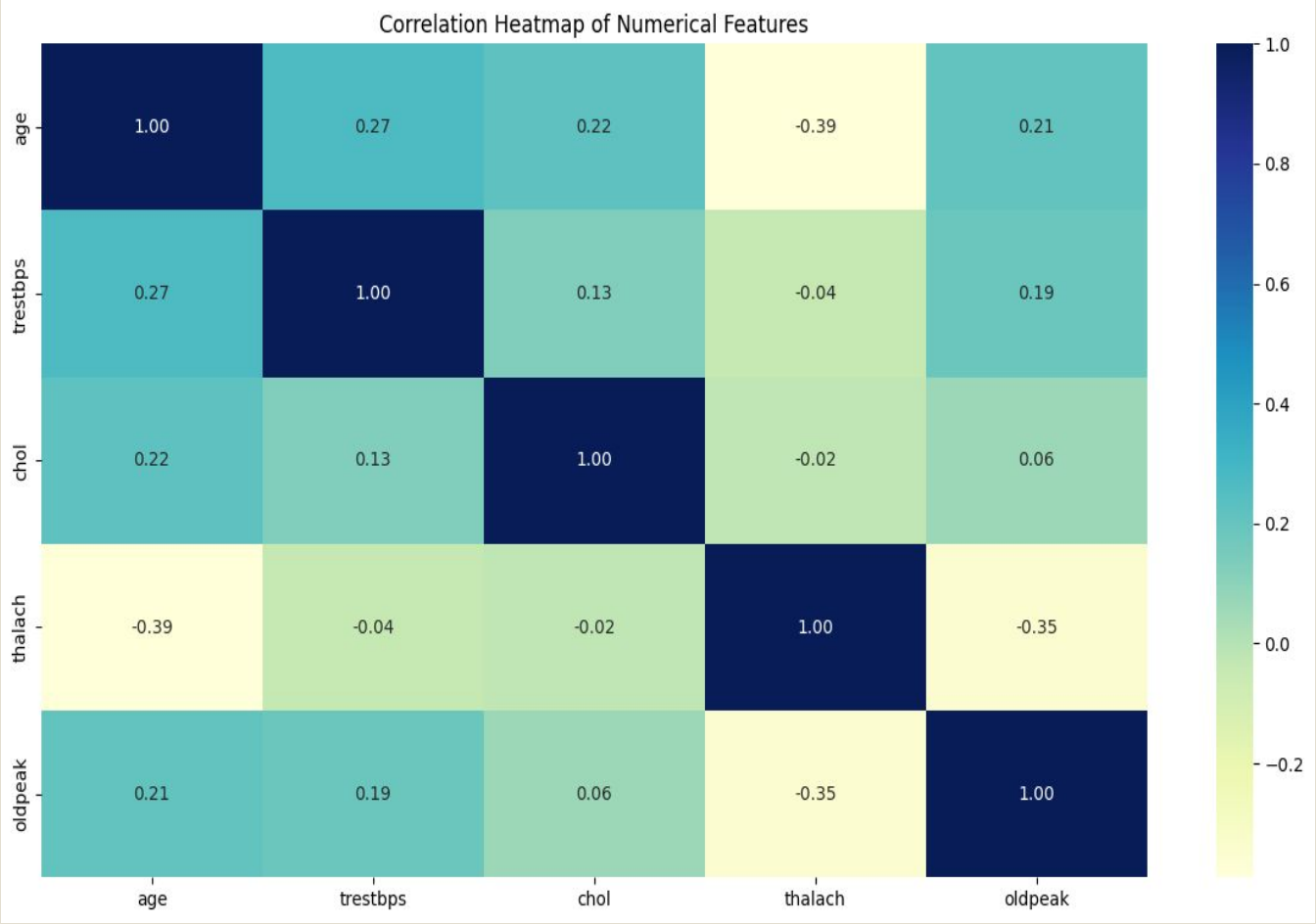
2.Skewed Distributions:

- Oldpeak is right-skewed (mean = 1.07, median = 0.8).
- Cholesterol shows high variance and possible outliers (mean > median).

3.Normal Distributions:

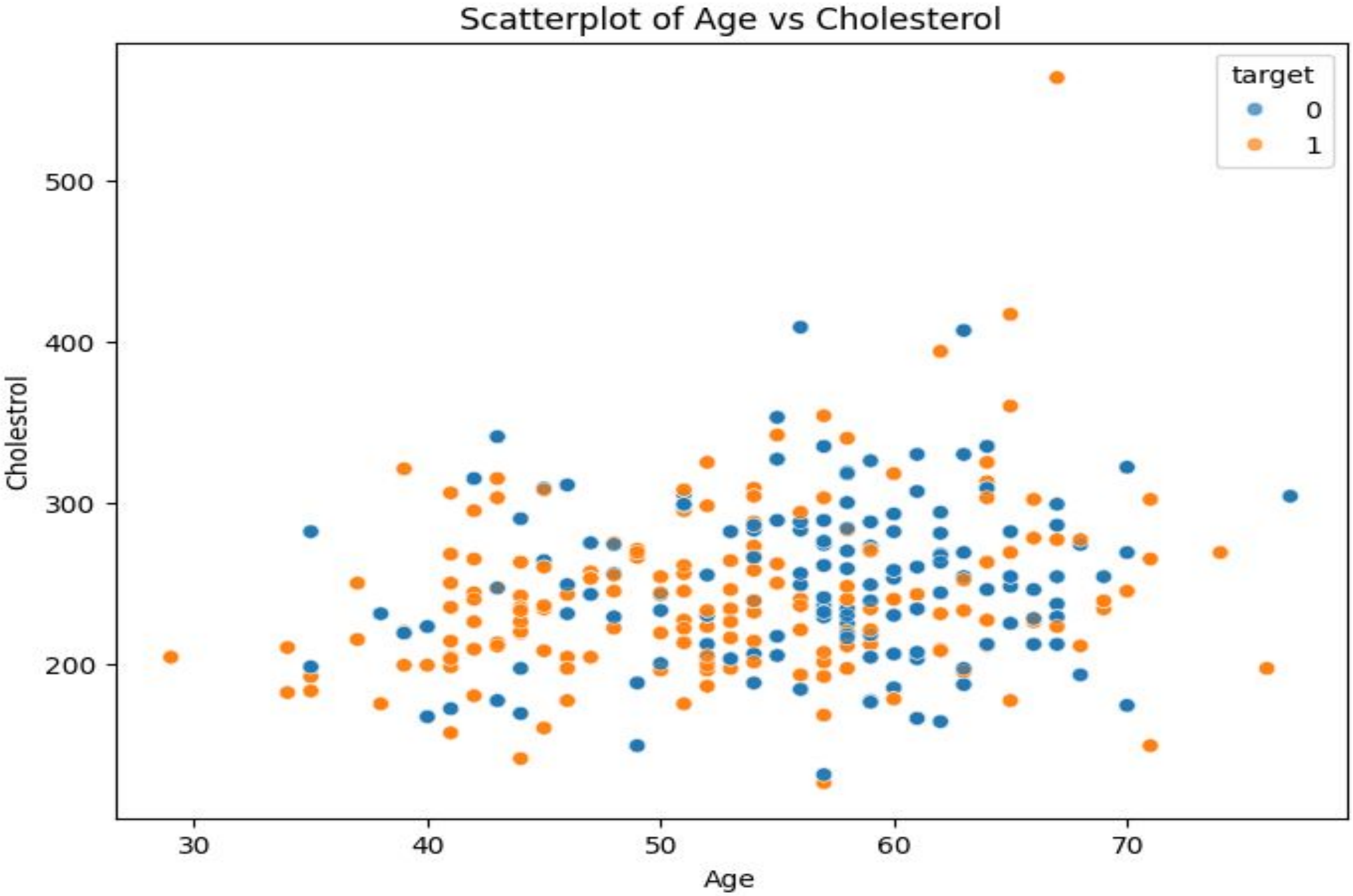
- Age is approximately normally distributed (mean = 54.43, median = 56).
- Most individuals fall within IQR of 48 to 61 years.

Correlation-Insights of Heatmap

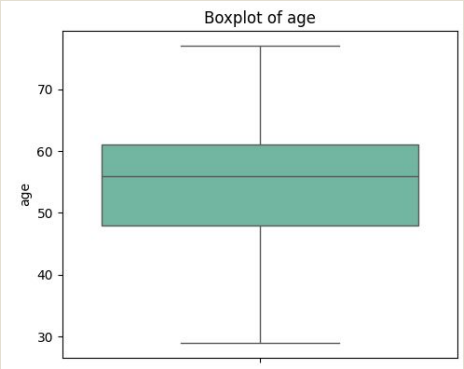


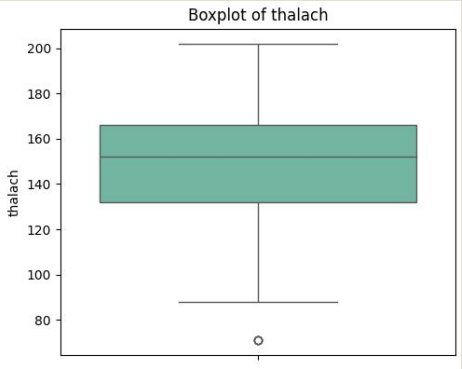
- **Age & Resting BP**: Weak positive correlation (0.27), suggesting older individuals tend to have slightly higher blood pressure.
- **Thalach (Max Heart Rate) & Age**: Negative correlation (-0.39), indicating heart rate tends to decrease with age.
- **Oldpeak & Thalach**: Negative correlation (-0.35), suggesting higher ST depression is linked to lower heart rate
- **Age & Cholesterol**: Weak positive correlation (~0.22 to 0.27), indicating slight increases in blood pressure and cholesterol with age.
- **Cholesterol**: Shows minimal correlation with other variables, indicating relatively independent behavior.

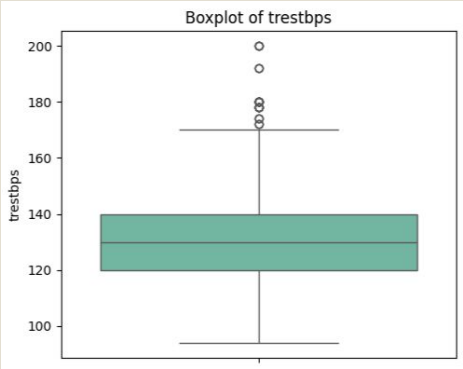
Multicollinearity: No strong correlations between most variables, reducing multicollinearity risk for modeling.

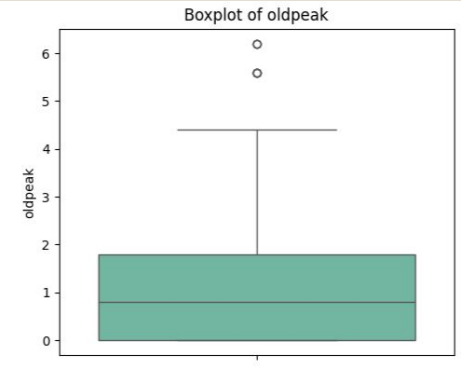

Scatterplot of Age vs Cholesterol
target
0
1
500
400
300
200
Cholestrol
30
40
50
60
70
Age

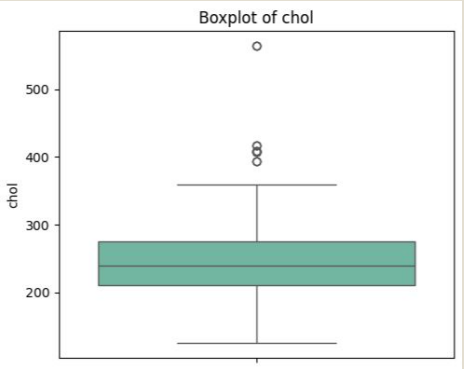
Removing Outliers











- **Age**: No significant outliers, distribution remains stable.
- **Trestbps (Resting Blood Pressure)**: Outliers above 180, likely due to high blood pressure. Removing them helps focus on typical values.
- **Chol (Cholesterol)**: Several outliers above 300. Removing them refines the focus on normal cholesterol levels.
- **Thalach (Max Heart Rate)**: Mild lower bound outlier. Removal may stabilize the distribution but has minimal impact.
- **Oldpeak (ST Depression)**: Outliers above 5, indicating extreme cases of ST depression. Removing them highlights typical responses.

Model Evaluation

- **False Positives (FP)**: Predicting heart disease when it's not present.
- **False Negatives (FN)**: Predicting no heart disease when it is present — critical for missed diagnoses and health complications.

Key Focus:

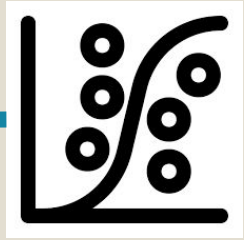
- **False Negatives (FN)** are the most dangerous, as they can result in undiagnosed heart disease, potentially leading to fatal consequences.

Reducing False Negatives:

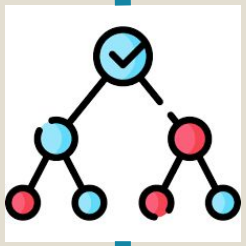
- **Prioritize Recall**: Aim to increase recall to capture high-risk patients. While higher recall might increase false positives, it's crucial to identify as many true cases as possible.
- **F1-Score**: If balancing both precision and recall is essential, the F1-score is the best metric, but recall should be prioritized in medical diagnoses.

BUILDING MODELS

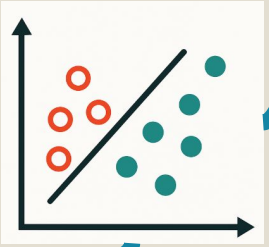
Predictive Models



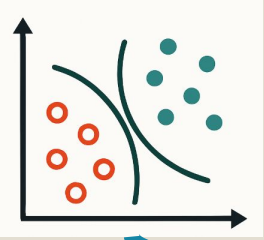
Logistic Regression



Decision Trees



Linear Discriminant Analysis



Quadratic Discriminant Analysis

Random Forest

Logistic Regression

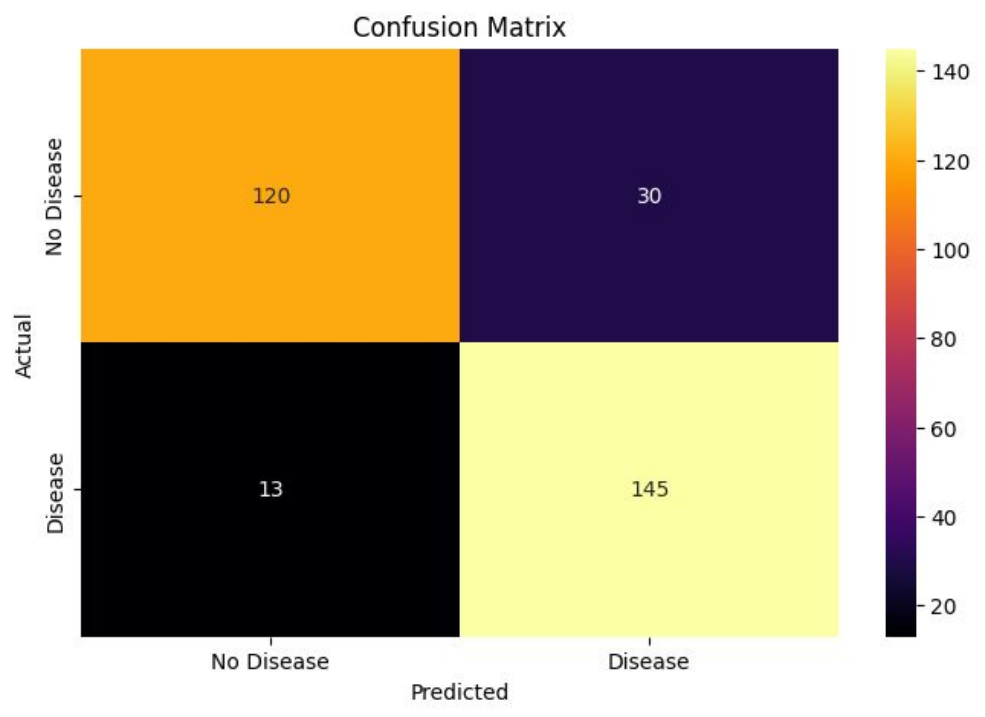
Binary Classification:

- The logistic regression model is used to predict the presence (1) or absence (0) of heart disease based on medical data.

Performance Overview:

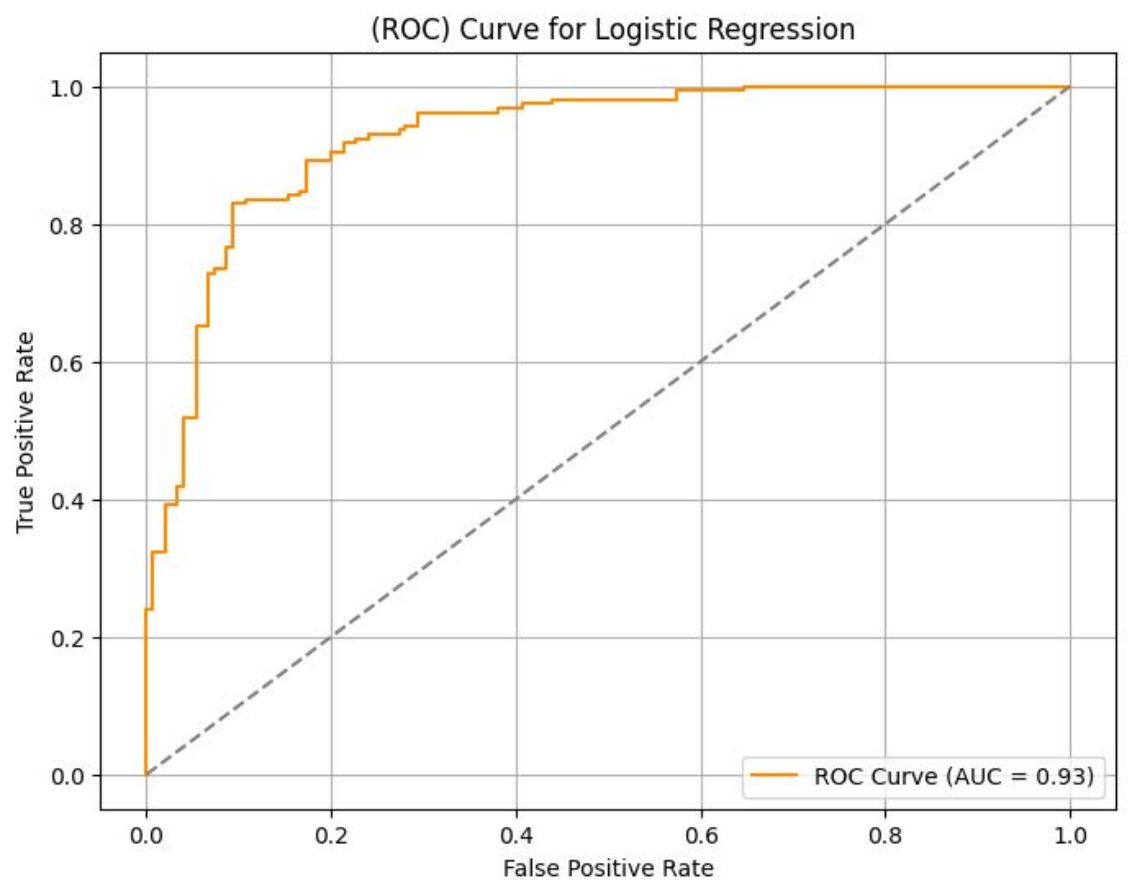
- **Train Accuracy**: 85%
- **Test Accuracy**: 84%
- This indicates **good generalization** with no overfitting. The model performs similarly on both training and testing datasets, implying it can handle unseen data well.

Key metrics derived (Testing):
Accuracy = 86%, Precision (Disease) = 83%, Recall (Disease) = 92%, Precision (No Disease) = 92%, Recall (No Disease) = 80%."



Confusion Matrix showing the classification results, including True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). This matrix helps evaluate the model's performance in predicting both classes. ***(Testing Data)***

ROC Curve - Logistic Regression



"ROC Curve with AUC = 0.93 indicating strong model performance and effectively differentiates between the two classes.

Stable F1-Score (~0.85): Balanced performance across both classes (No Disease and Disease).

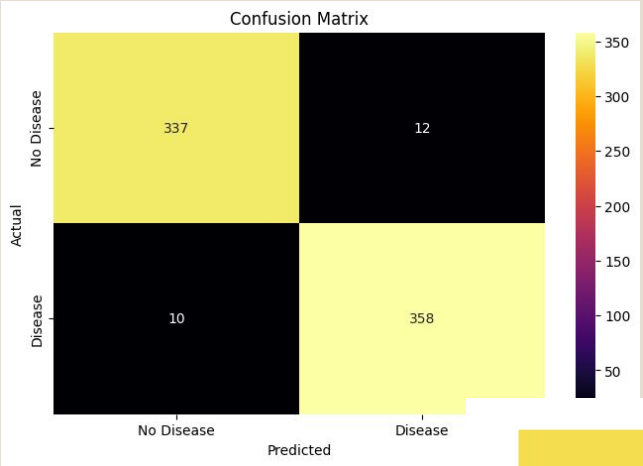
Class Distribution:

- **Class 0 (No Disease)**: 150 samples
- **Class 1 (Heart Disease)**: 158 samples
  - The dataset is well-balanced with minimal class imbalance.

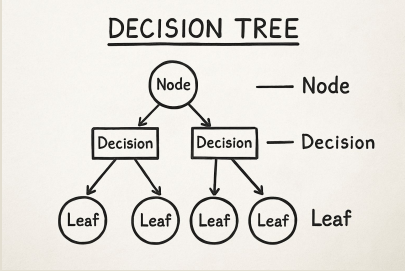
Test Performance: Higher precision and recall on test data than on training data, suggesting the model may be slightly underfitting but is well-regularized.

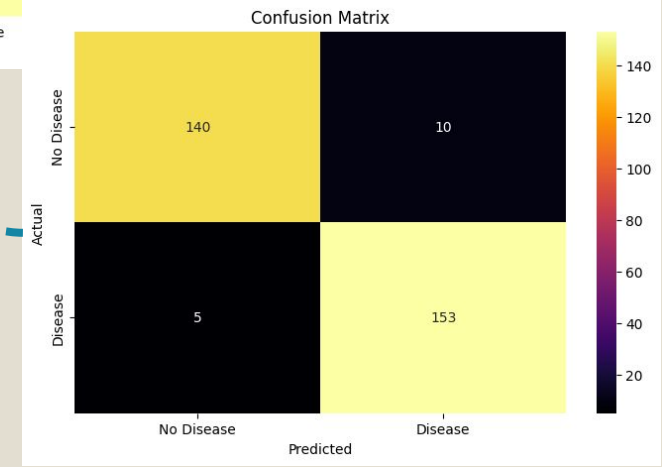
Potential Improvement: Consider **threshold tuning** to optimize the tradeoff between precision and recall.

Tree Induction - Decision Trees



(Training)





Key metrics derived (Testing):
Accuracy = 86%, Precision (Disease) = 83%, Recall (Disease) = 92%, Precision (No Disease) = 92%, Recall (No Disease) = 80%.

Tree Induction:
A method for building decision trees by learning rules from data to split it into subgroups for classification or regression.

How it Works:

- Starts at the **root node** with the entire dataset.
- **Splits** data based on features using metrics like **information gain**.
- Recursively grows branches until stopping criteria are met.
- **Optional pruning** helps prevent overfitting.

Performance:

- **Train Accuracy**: 97%
- **Test Accuracy**: 95%

High predictive performance with strong precision, recall, and F1-scores across both classes.

ROC Curve - Decision Tree

Excellent Heart Disease Detection:

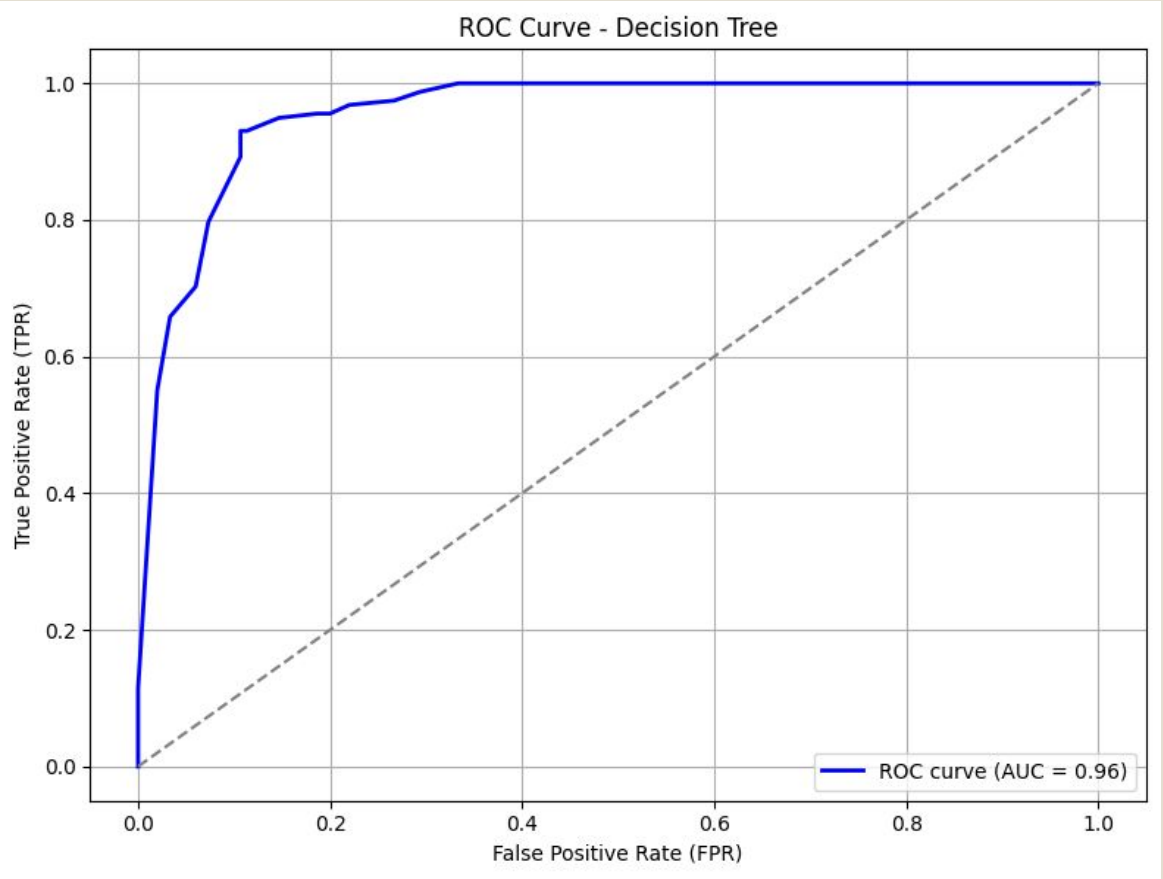
- **Class 1 recall**: 97% on both train and test sets.
- This **greatly minimizes false negatives** and ensures reliable identification of at-risk patients.

Balanced Complexity:

- **Depth capped at 6** to control overfitting.
- Achieves a good balance between model complexity and generalization.

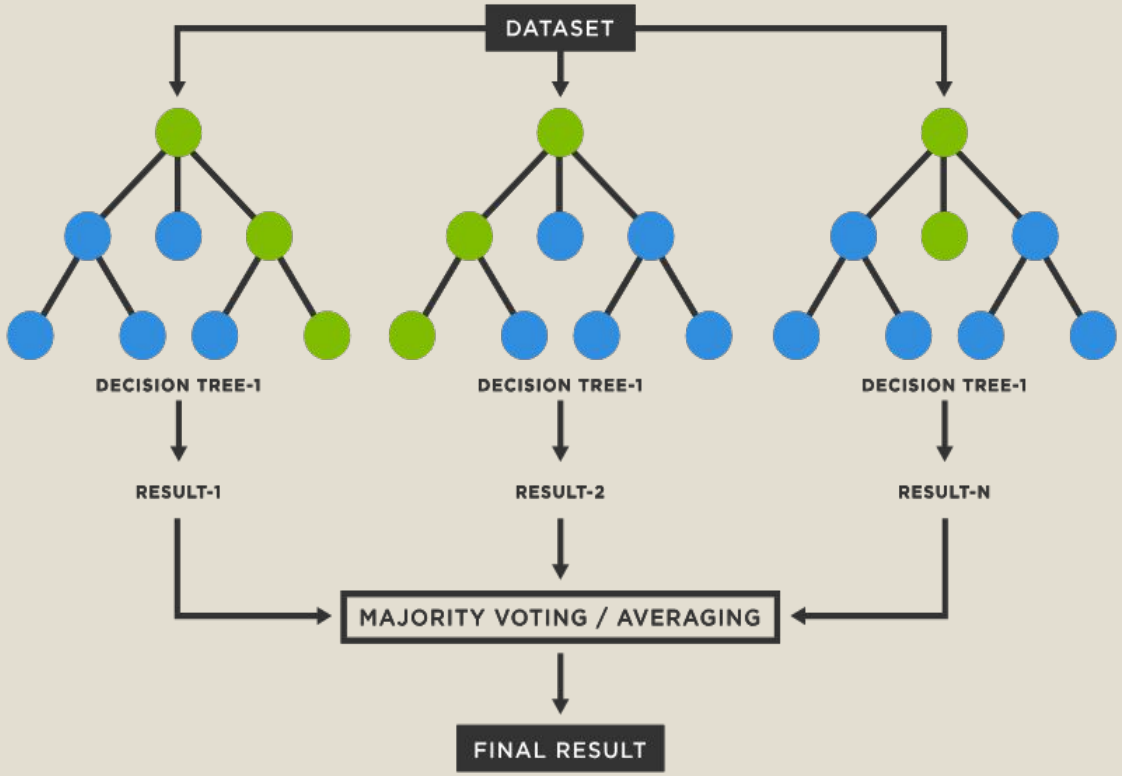
Strong Model Performance:

- The **ROC Curve** with **AUC = 0.98** demonstrates **superior class separation**, with minimal false positives and false negatives



ROC Curve with AUC = 0.98 showing strong model performance and the ability to effectively distinguish between classes.

RANDOM FOREST



What is Random Forest?

- An ensemble method that builds multiple decision trees and averages their predictions for improved accuracy and stability.

Why Use It for Heart Disease Prediction?

- Handles complex, non-linear relationships between health indicators (age, cholesterol, BP, etc.) and heart disease risk.
- Reduces overfitting compared to a single decision tree.
- Offers high predictive power and robustness.

Strengths:

- Captures feature interactions
- Works well with both categorical and numerical variables

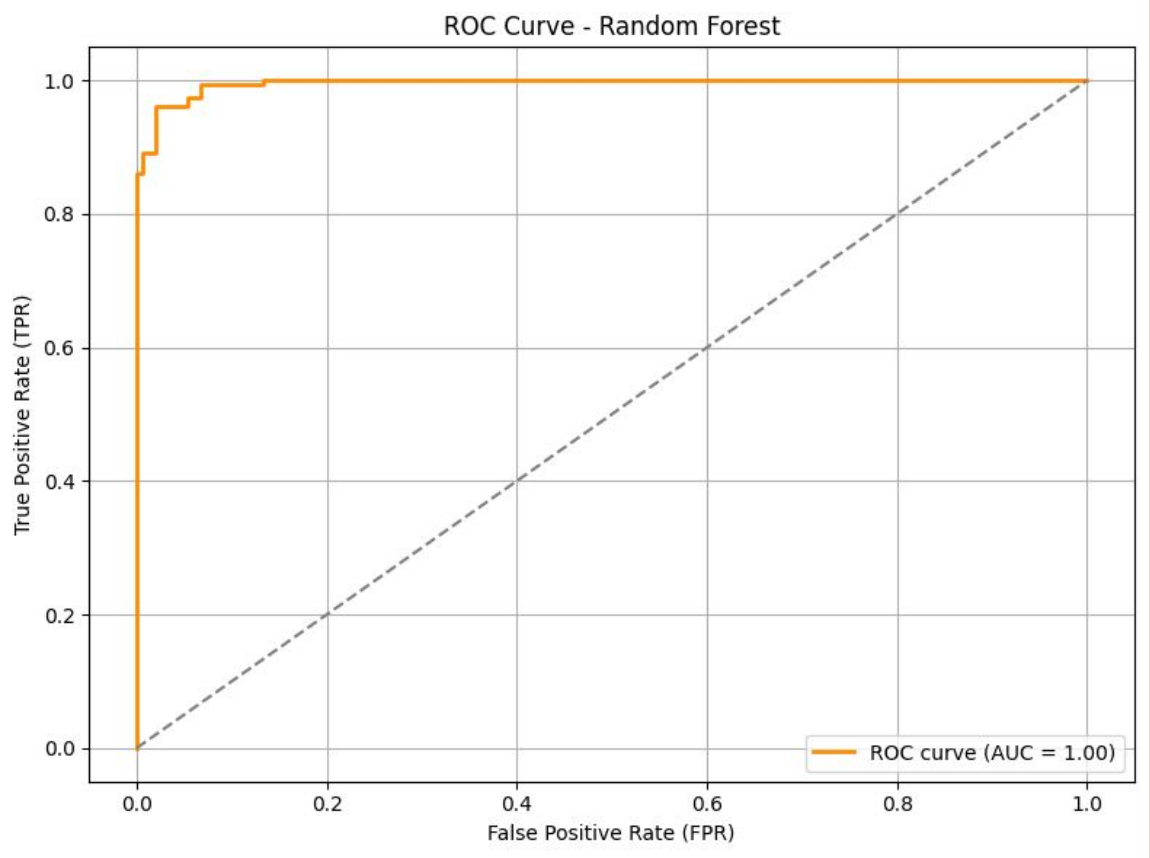
Random Forest- Takeaways

High Accuracy (96%):
- Correctly classifies most test samples with excellent overall performance.

Balanced Precision & Recall:
- Class 0 (No Disease): Precision = 0.99, Recall = 0.93
- Class 1 (Disease): Precision = 0.94, Recall = 0.99
- Reliable across both classes.

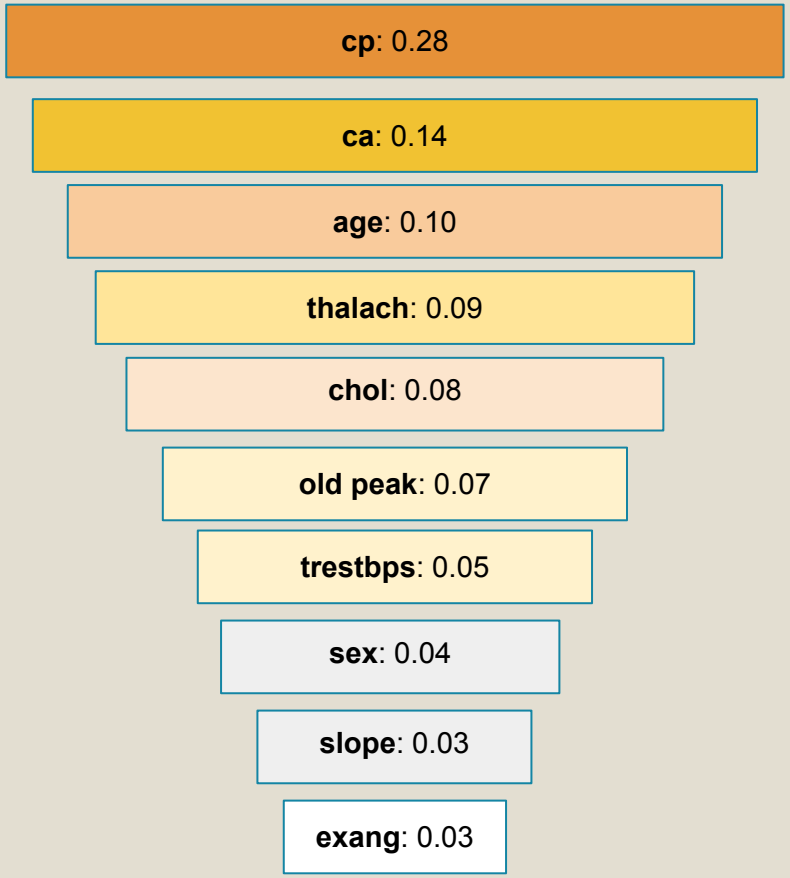
Low Misclassification:
- Only 10 false positives and 2 false negatives
- Strong at detecting true positives with minimal false alarms.

ROC-AUC Score: ~1.00
- Indicates near-perfect distinction between diseased and non-diseased cases.



*The Random Forest classifier demonstrates an exceptional **AUC of approximately 1.00**, indicating near-perfect separation between heart disease and non-disease cases. While not exactly 1.00, the ROC curve performance is extremely close, reflecting highly reliable predictions with minimal false positives or negatives.*

Feature Importance



- **Chest Pain Type (cp)**:
  Most important; indicates the level of heart stress.
- **Blocked Vessels (ca)**:
  More blocked vessels increase heart disease risk.
- **Age (age)**:
  Older age significantly raises the likelihood of heart disease.
- **Maximum Heart Rate (thalach)**:
  Lower heart rate during exercise indicates heart issues.
- **Cholesterol Level (chol)**:
  High cholesterol contributes to plaque buildup in arteries.
- **Old Peak (old peak)**:
  Depression of ST segment during exercise indicates heart stress.
- **Resting Blood Pressure (trestbps)**:
  Higher resting blood pressure is a risk factor for heart disease.
- **Sex (sex)**:
  Gender impacts heart disease risk, with men generally at higher risk.
- **Slope (slope)**:
  Slope of the peak exercise ST segment is a smaller factor in risk.
- **Exercise-Induced Angina (exang)**:
  Angina during exercise indicates heart stress, but has lower importance.

Linear Discriminant Analysis

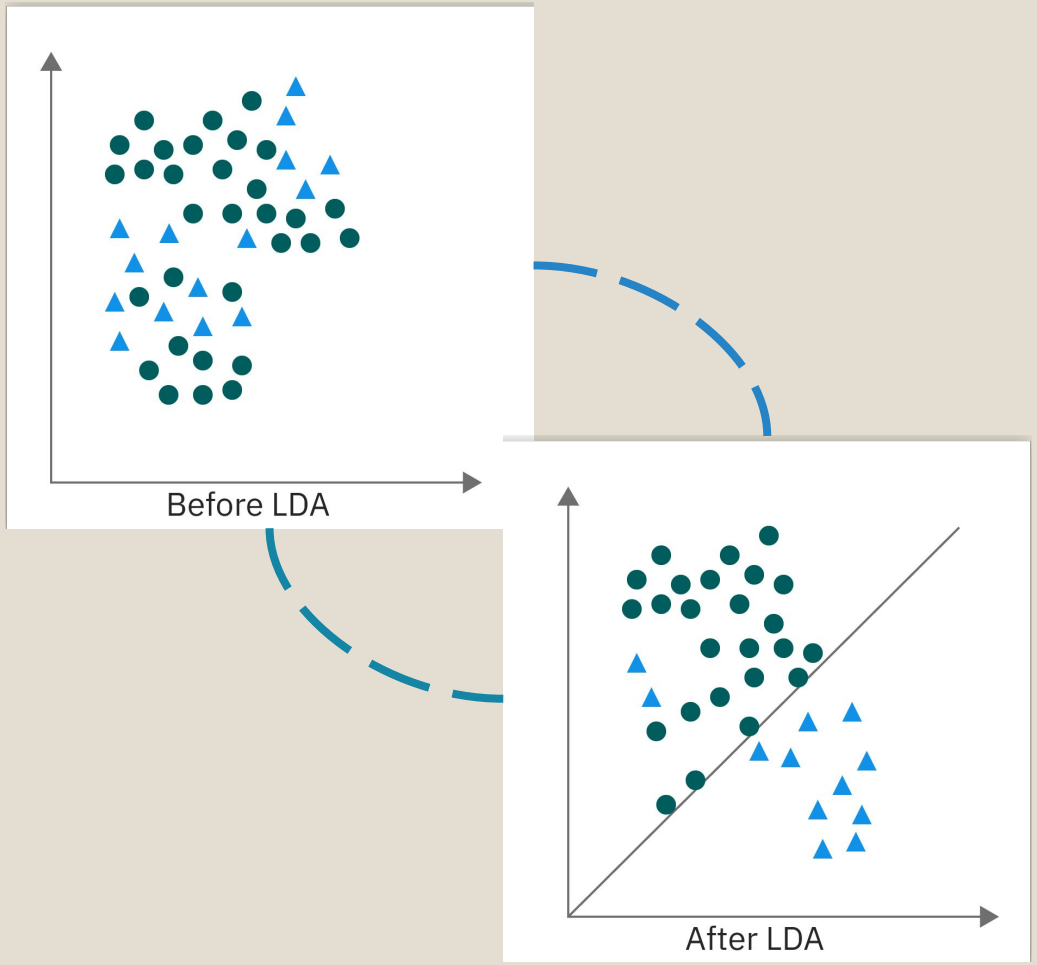
What is LDA?
A supervised learning technique used for **classification** and **dimensionality reduction**, especially in **multi-class problems**.

How It Works:

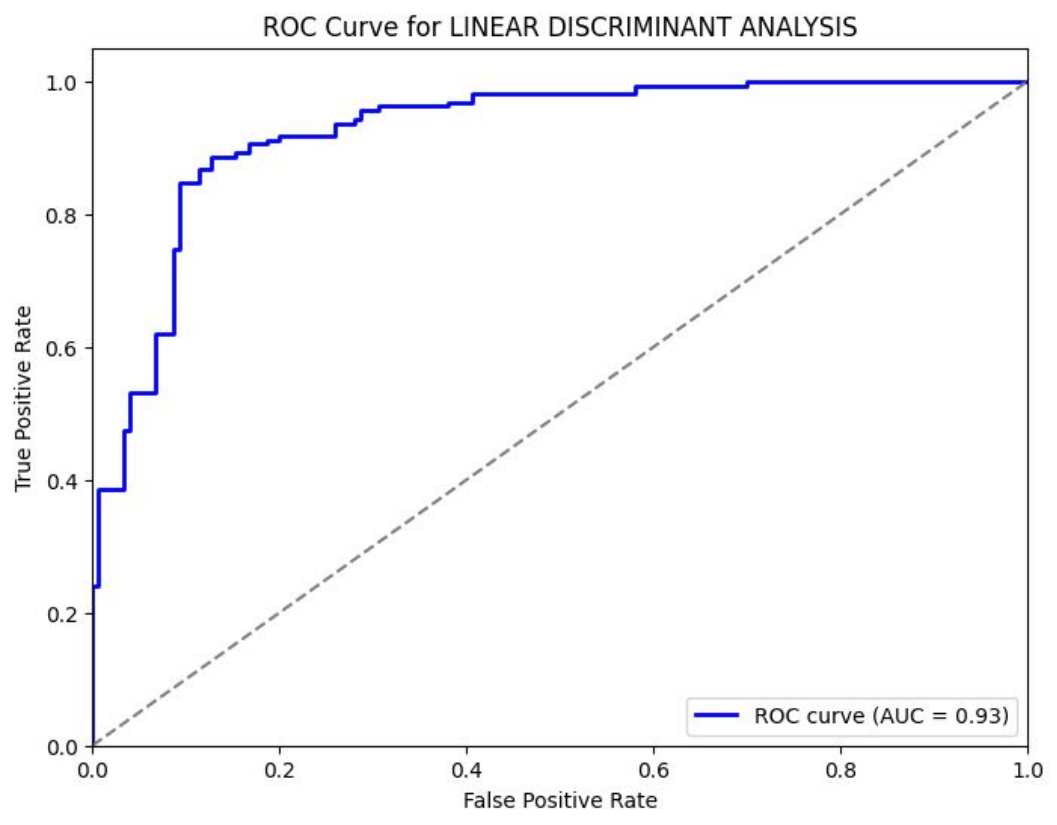
- **Compute class means** for each category
- **Construct scatter matrices** (within-class & between-class)
- **Maximize separation** by finding a linear combination of features
- **Project data** onto new axes for clear decision boundaries

Why use LDA?

- Great for linearly separable classes
- Helps reduce dimensionality



Linear Discriminant Analysis- Results



*The **LDA ROC curve** demonstrates a high **AUC of 0.93**, indicating strong classification performance. The curve closely resembles that of **Logistic Regression**, suggesting similar predictive capabilities.*

High Recall (0.92) for Class 1:

- Effectively detects heart disease and reduces false negatives

Balanced Metrics:

- Precision and F1-scores are consistent across both classes

Strong Generalization:

- Training Accuracy: 0.83 | Test Accuracy: 0.84
- Low risk of overfitting

High AUC (0.93):

- ROC curve shows strong class separation Well-suited for linearly separable data

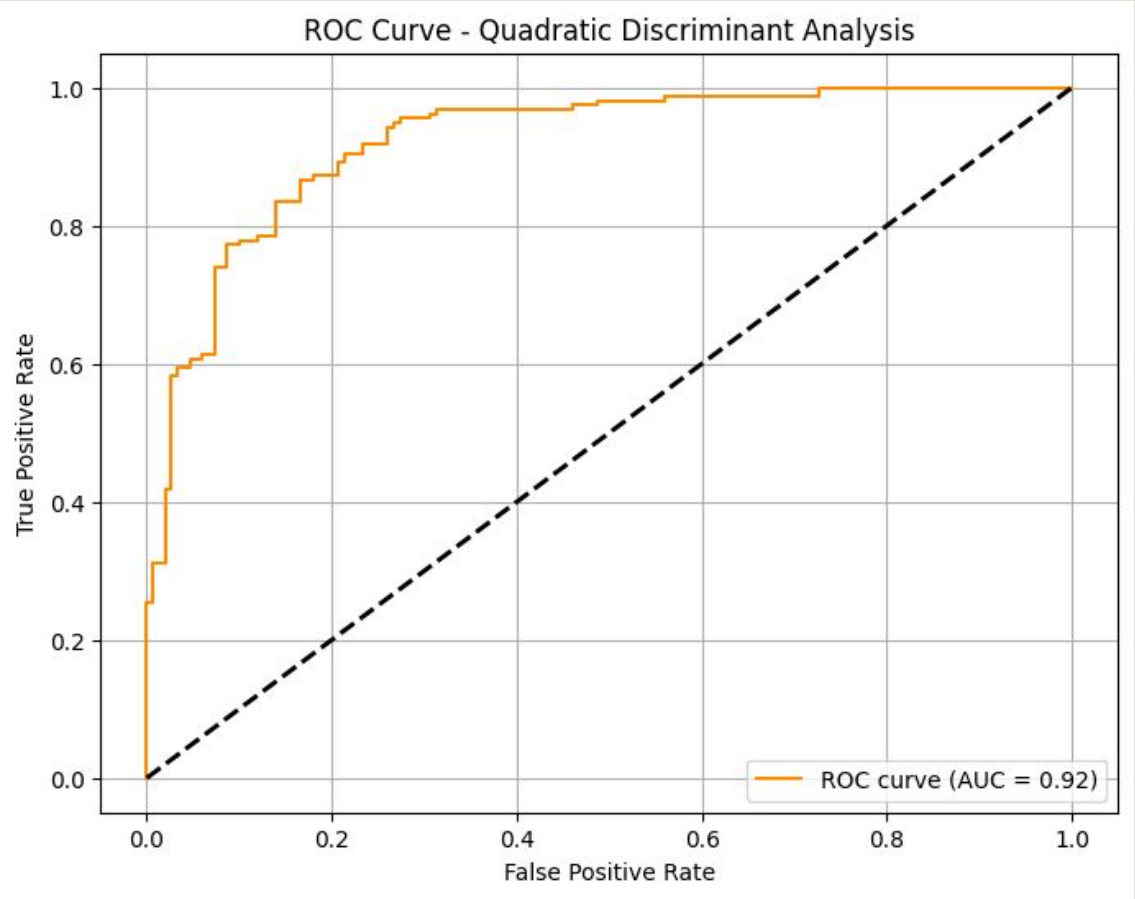
Quadrant Discriminant Analysis

Non-linear Decision Boundaries:
Uses class-specific covariance matrices, ideal when class distributions are non-linearly separable.

High Sensitivity to Positive Class:
Recall = 0.91 (train), 0.87 (test) — effectively identifies heart disease cases, minimizing false negatives.

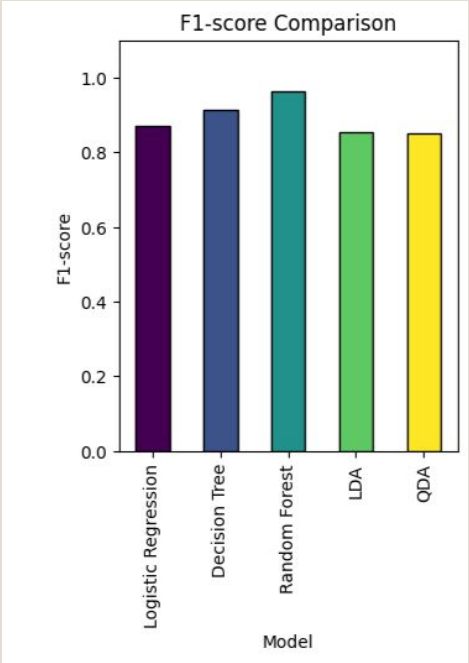
Balanced Performance:
Precision, Recall, and F1-scores are consistent across both classes. **Accuracy** = 0.88 (train), 0.84 (test).

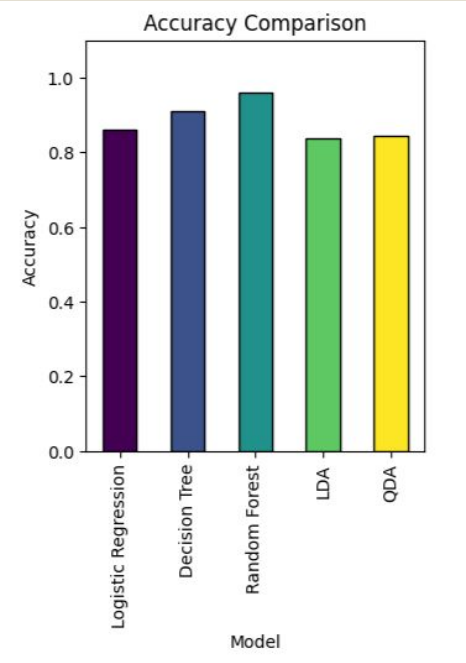
Strong Generalization:
Minimal performance drop from training to test, indicating low overfitting and good reliability on unseen data.

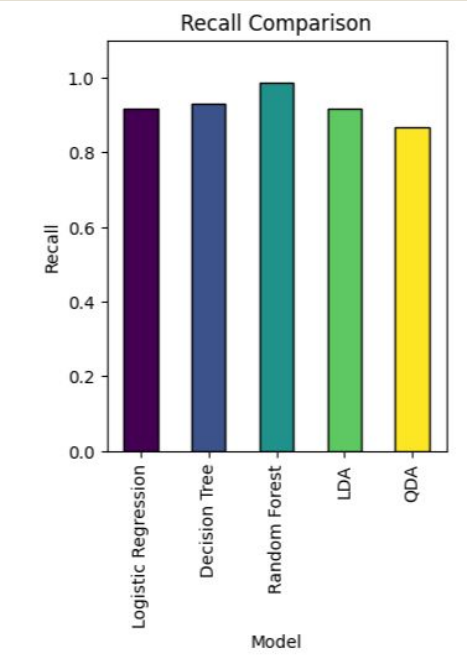


ROC Curve for QDA (AUC = 0.92) *— Demonstrates strong class separation and model effectiveness in distinguishing between disease and no disease.*

Model Comparison







Random Forest consistently shows top performance across all three metrics (**F1-score**, **Accuracy**, and **Recall**), making it the best model for **heart disease prediction**.

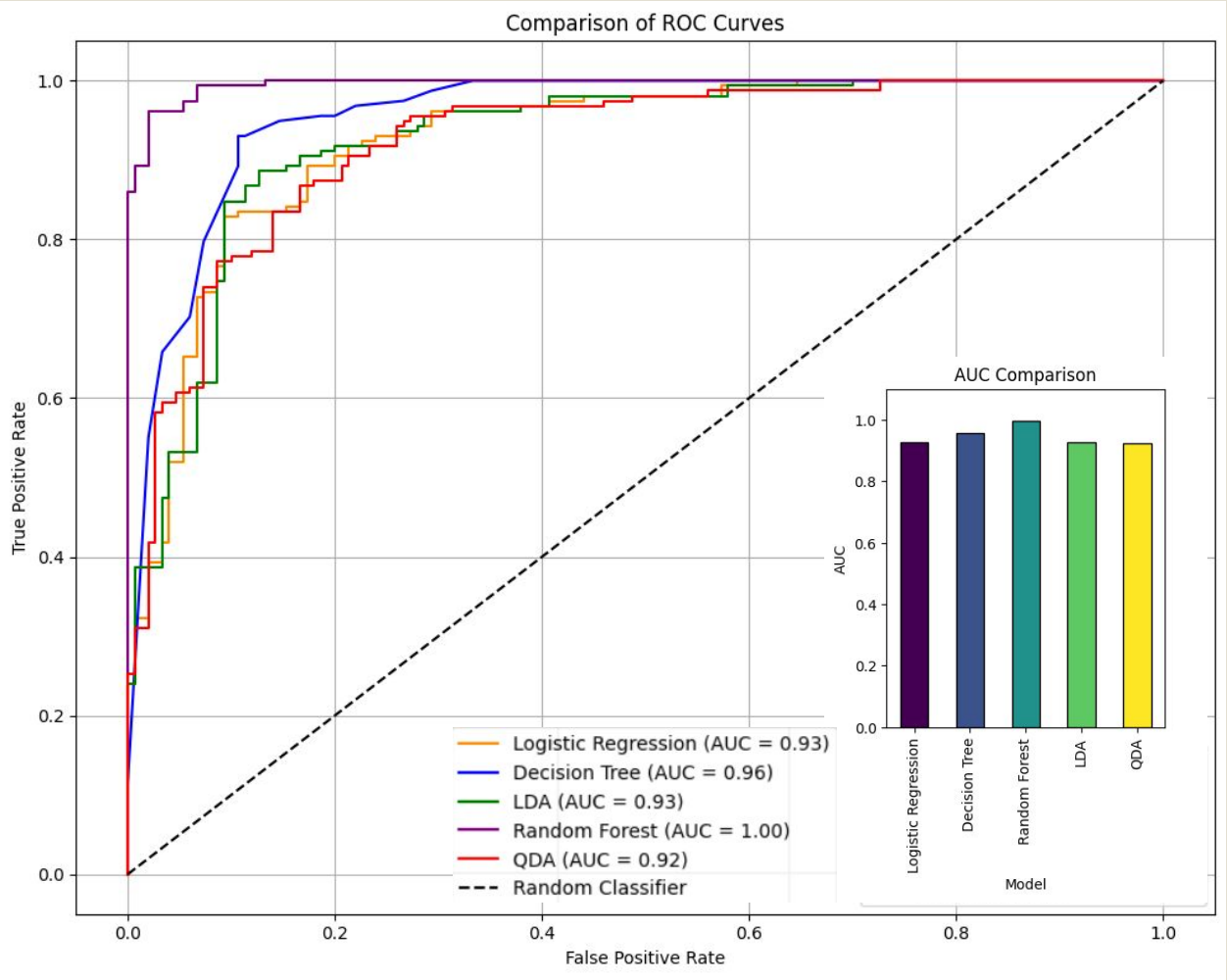
Decision Tree and **Logistic Regression** are also strong contenders, performing similarly across the metrics.

F1-score:
Random Forest and **Decision Tree** lead, followed by **Logistic Regression** and **LDA**. **QDA** has the lowest F1-score.

Accuracy:
Random Forest achieves the highest accuracy, with **Logistic Regression** and **Decision Tree** following. **QDA** has the lowest accuracy.

Recall:
Random Forest excels in recall, followed by **Logistic Regression** and **Decision Tree**. **QDA** performs the weakest in recall.

Model comparison



ROC Curve:

- **Random Forest** outperforms all models, with the highest True Positive Rate and lowest False Positive Rate.
- **Decision Tree** and **LDA** perform similarly, followed by **Logistic Regression** and **QDA**.
- The **Random Classifier** serves as a baseline with an AUC of 0.5

AUC Values:

- **Random Forest**: AUC ~ 1.0 (best performance).
- **Decision Tree**, **Logistic Regression**, **LDA**: AUC ≈ 0.93.
- **QDA**: AUC = 0.92 (still strong).

Conclusion: **Random Forest is the top performer, followed by Decision Tree and LDA. QDA performs slightly worse but is still competitive.**

In the context of **Heart disease prediction**, **Random Forest** provides the best accuracy and generalization, making it the most suitable model for this task.

Business Recommendations

	Recommendation	Implementation
1	Identify High-Risk Patients Early	Train Random Forest model using historical patient data for risk prediction.
2	Personalize Care Plans	Generate risk scores for each patient and integrate with EHR systems.
3	Optimize Use of Doctors and Diagnostic Tools	Build real-time prioritization system based on patient risk scores.
4	Help Insurance Companies Assess Risk	Use Random Forest to predict heart disease risk and adjust premiums.
5	Create Preventive Health Programs	Identify key risk factors and design targeted preventive programs.

Thank You